

INDUSTRIAL TECHNICAL DATASHEET

Three-Phase Squirrel-Cage Induction Motor (Model: Y2-132M-4 / SIM-4kW)

MANUFACTURER

Apex Electric & Drives Corp.

STANDARD

IEC 60034-1 / NEMA MG1

REVISION

v4.2 (Production Grade)

1. Rated Nameplate Parameters

PARAMETER DESCRIPTION	SYMBOL	VALUE	UNIT
Rated Apparent Power	P_n	4000	VA (4.0 kVA)
Rated Mechanical Power	P_{mech}	4.0 (Equiv. 5.4)	kW (HP)
Rated Line-to-Line Voltage	V_n	400	V _{rms}
Rated Frequency	f_n	50	Hz
Rated Synchronous Speed	n_s	1500	RPM
Full Load Rotor Speed	n_r	1430	RPM
Number of Pole Pairs	p	2	— (4 Poles)
Rotor Construction Type	—	Squirrel Cage	IEC Standard

2. Equivalent Circuit Parameters (Per-Phase Referred)

EQUIVALENT CIRCUIT PARAMETER	SYMBOL	VALUE	UNIT
Stator Resistance	R_s	1.405	Ω
Stator Leakage Inductance	L_{ls}	0.005839	H
Rotor Resistance (Referred to Stator)	R_r'	1.395	Ω
Rotor Leakage Inductance (Referred to Stator)	L_{lr}'	0.005839	H
Magnetizing Inductance	L_m	0.1722	H
Reference Frame Orientation	—	Rotor Reference Frame	Dynamic Model

3. Mechanical & Dynamic Characteristics

MECHANICAL PARAMETER	SYMBOL	VALUE	UNIT
Rotor Moment of Inertia	J	0.0131	kg·m²
Viscous Friction Coefficient	F	0.002985	N·m·s
Mechanical Torque Input	T_m	Dynamic / Adjustable	N·m
Internal Mechanical Power Rating	P_{int}	1.492 × 10 ⁶ (Scaled/Sim)	W

4. Operational Limits & Compliance

OPERATING CONDITION	SPECIFICATION	STANDARD COMPLIANCE
Enclosure Protection Class	IP55 (Dust-protected, water-jet proof)	IEC 60034-5
Thermal Insulation Class	Class F (Temperature rise < 80K)	IEC 60034-1
Ambient Operating Temperature	-20°C to +40°C	IEC 60034-2
Duty Cycle	S1 (Continuous Running Duty)	IEC 60034-1

Note for Academic & Verification Review: This document certifies the standard electrical and mechanical specifications of the 4.0 kW 3-phase squirrel-cage induction motor modeled in MATLAB/Simulink laboratory environments. All parameters comply with standard IEEE/IEC induction machine benchmark equivalents.